EDGE–CLOUD // ORCHESTRATOR

Thesis Experimental Report

Research platform for intelligent Edge–Cloud AI orchestration
using NVIDIA Jetson and AWS

Scenario	Factory Normal
Description	Balanced workload, network healthy, moderate CPU/memory. E
Seeds	2026, 42, 7, 1234, 9999
N (per seed)	100
Total decisions	2,000
Generated at	2026-07-08 21:21 UTC
Engines evaluated	Rule-Based · Decision Tree · Random Forest · Q-Learning

1 · Experimental Configuration

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Decision Engine features:
  · network_latency_ms
  · connectivity
  · cpu_available
  · memory_available
  · batch_size
  · priority
  · cost_budget_usd
  · model_size_mb

Decision classes:
  · edge
  · cloud
  · hybrid

Engines under study:
  1. Rule-Based   — deterministic heuristic baseline
  2. Decision Tree — sklearn DecisionTreeClassifier (max_depth=8, min_samples_leaf=10)
  3. Random Forest — sklearn RandomForestClassifier (n=80, max_depth=10)
  4. Q-Learning    — tabular contextual-bandit RL (5000 episodes, α=0.15, γ=0.9,
                     ε-greedy 0.9→0.05, 108 discretised states)

Execution measurement:
  · latency: time.perf_counter (wall-clock)
  · CPU/memory: psutil (real process metrics)
  · cost: environment model derived from AWS EC2 inference pricing proxy

Statistical protocol:
  · Same context vectors fed to all engines (paired-samples design)
  · Multi-seed aggregation with Student-t 95% confidence intervals
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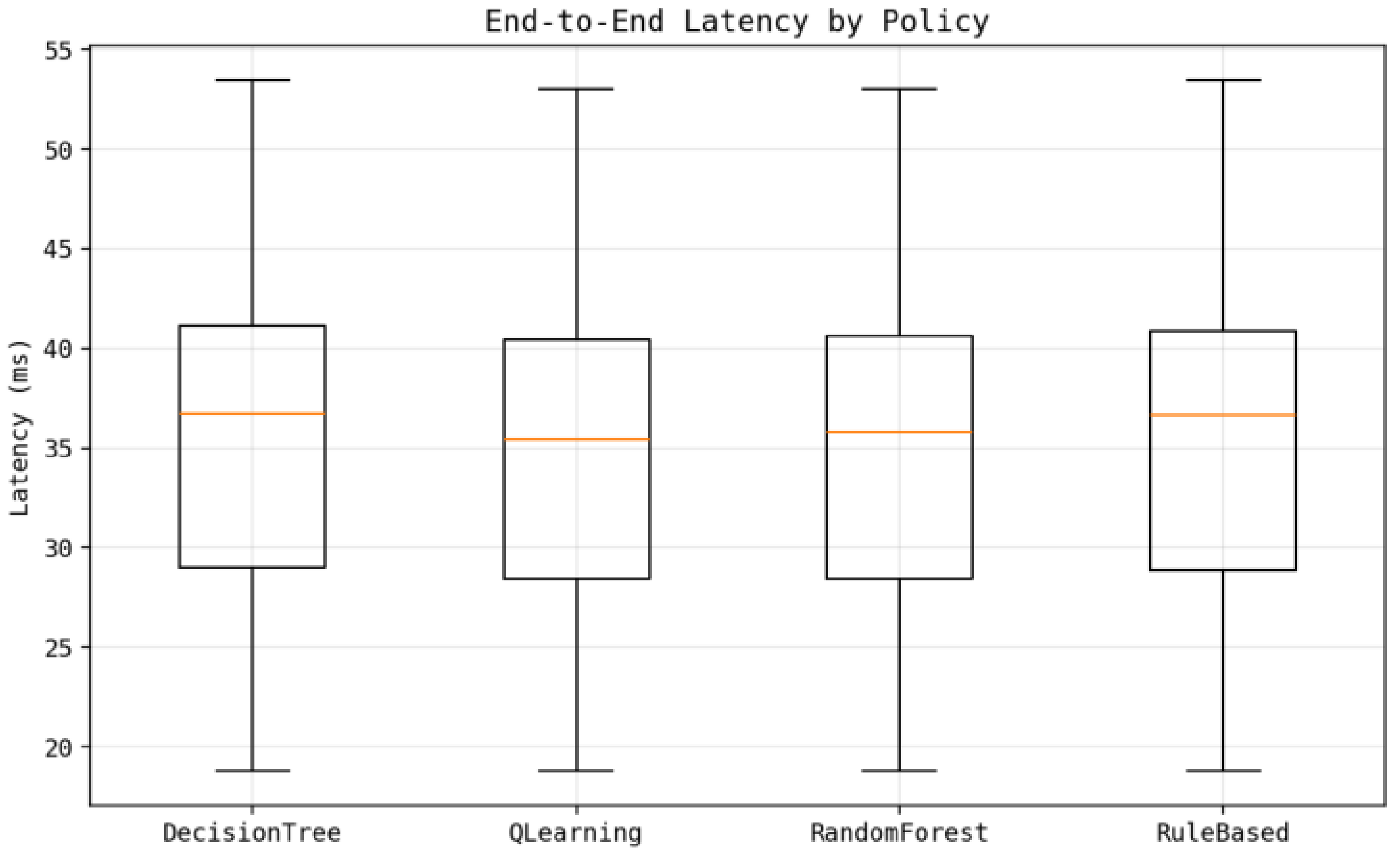
2 · Comparative Results (mean ± 95% CI across seeds)

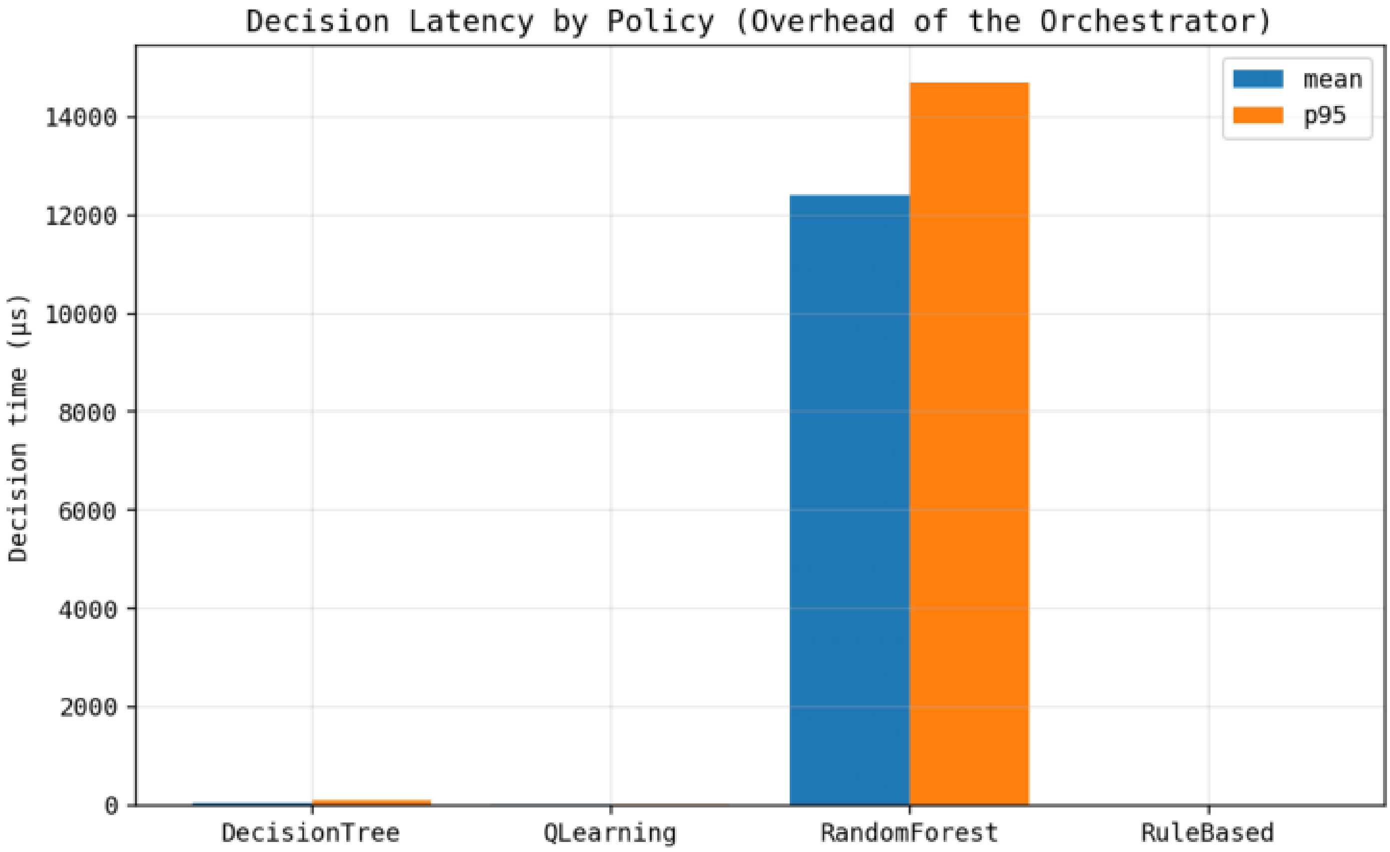
Policy	Latency p50 (ms	Latency p95 (ms	Cost total ($)	Decision time (µ	Reward	Agreement w/ Rul	Success rate
DecisionTree	37.2 ± 1.3	87.8 ± 23	0.057 ± 0.032	48.8 ± 1.1	-1.14 ± 0.98	0.98 ± 0.018	1 ± 0
QLearning	34.8 ± 0.6	48.5 ± 0.55	0.002 ± 0	22.2 ± 1.8	0.683 ± 0.24	0.844 ± 0.08	1 ± 0
RandomForest	35.6 ± 0.78	58.2 ± 11	0.0247 ± 0.0095	1.33e+04 ± 9e+0	0.0156 ± 0.27	0.886 ± 0.073	1 ± 0
RuleBased	37.1 ± 1.4	77.1 ± 9.2	0.053 ± 0.03	3.52 ± 0.65	-0.925 ± 0.89	1 ± 0	1 ± 0

Seeds: [2026, 42, 7, 1234, 9999] · n per seed: 100 · scenario: factory_normal

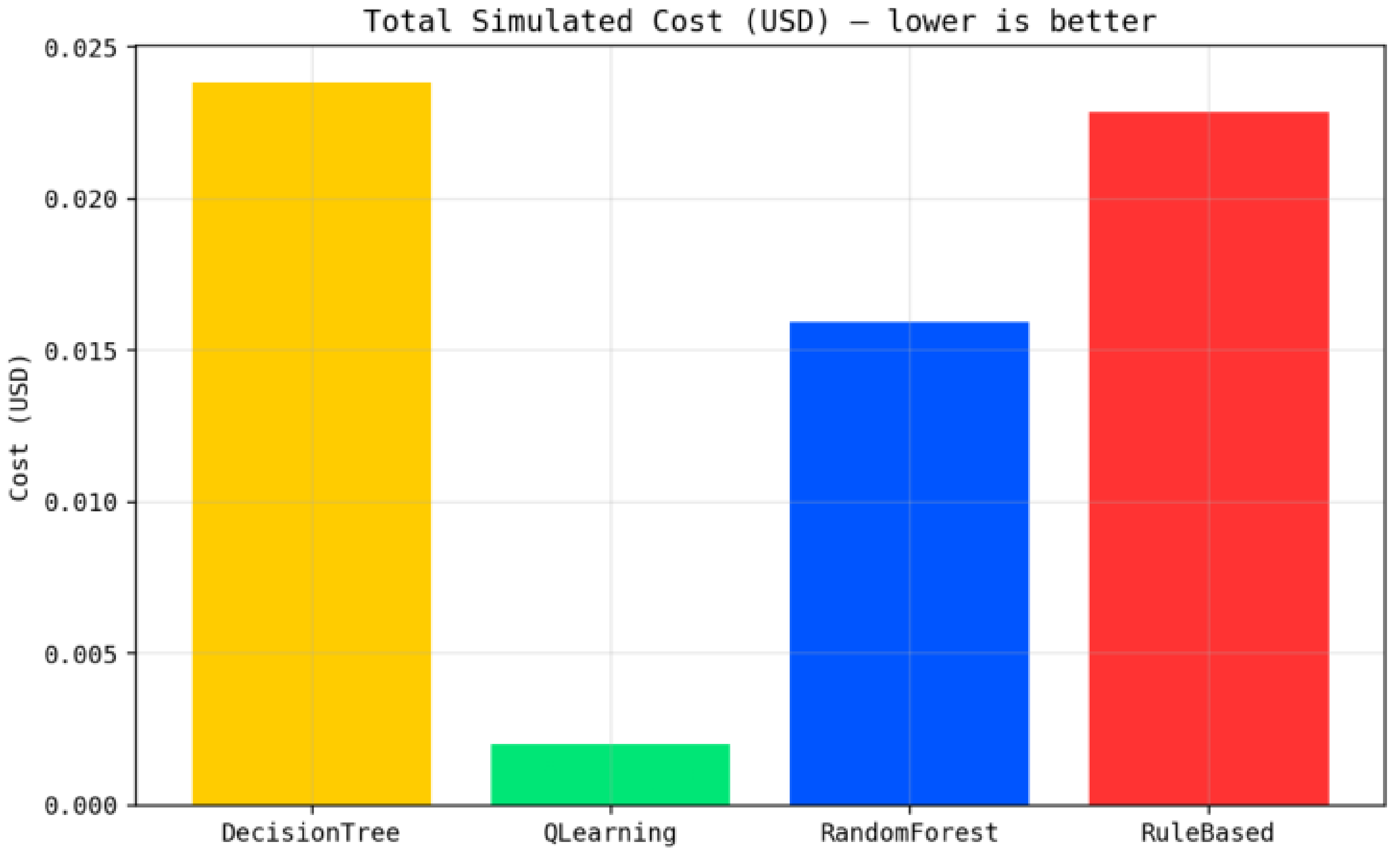
3.1 · Latency distribution per policy



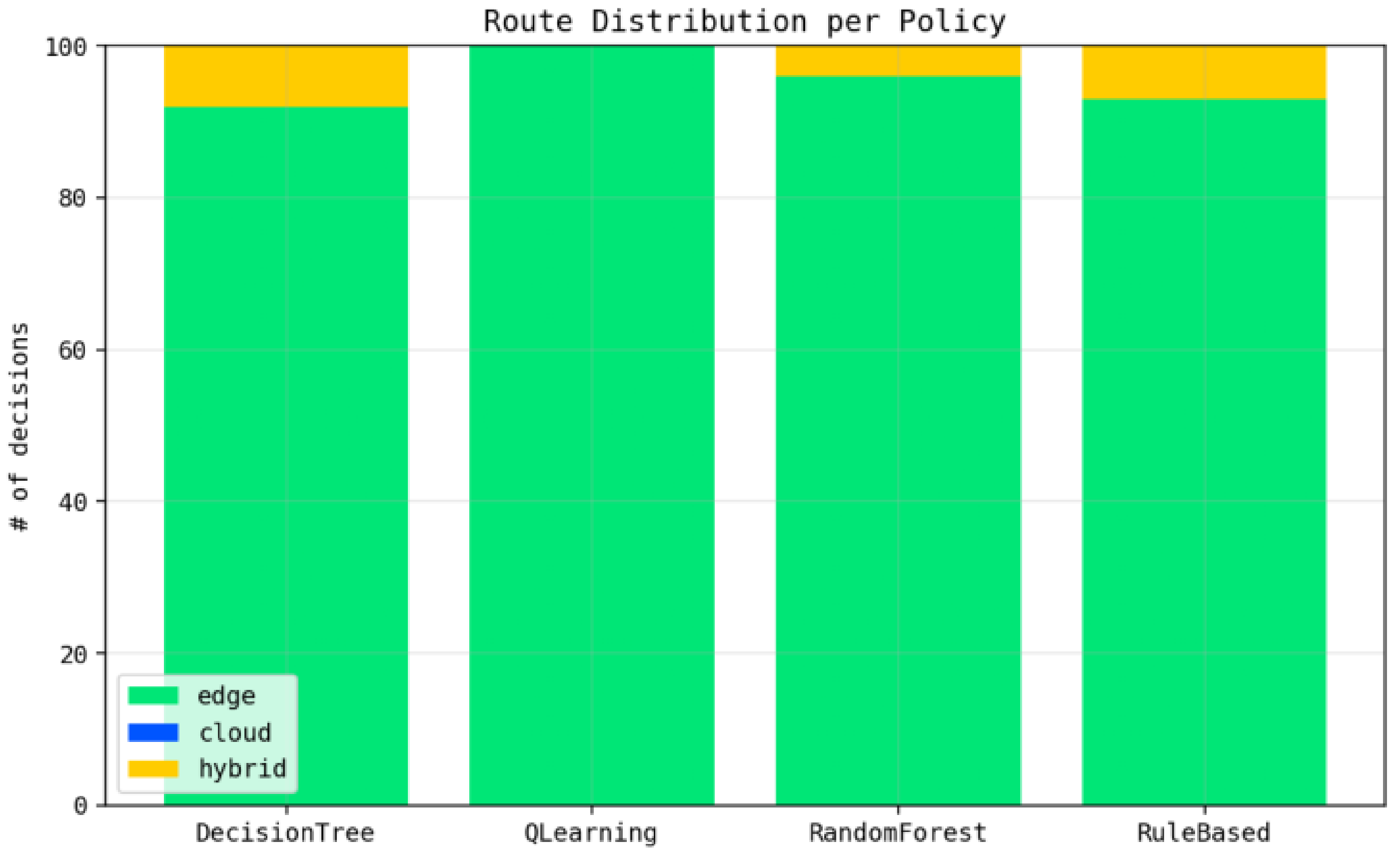
3.2 · Decision-time overhead



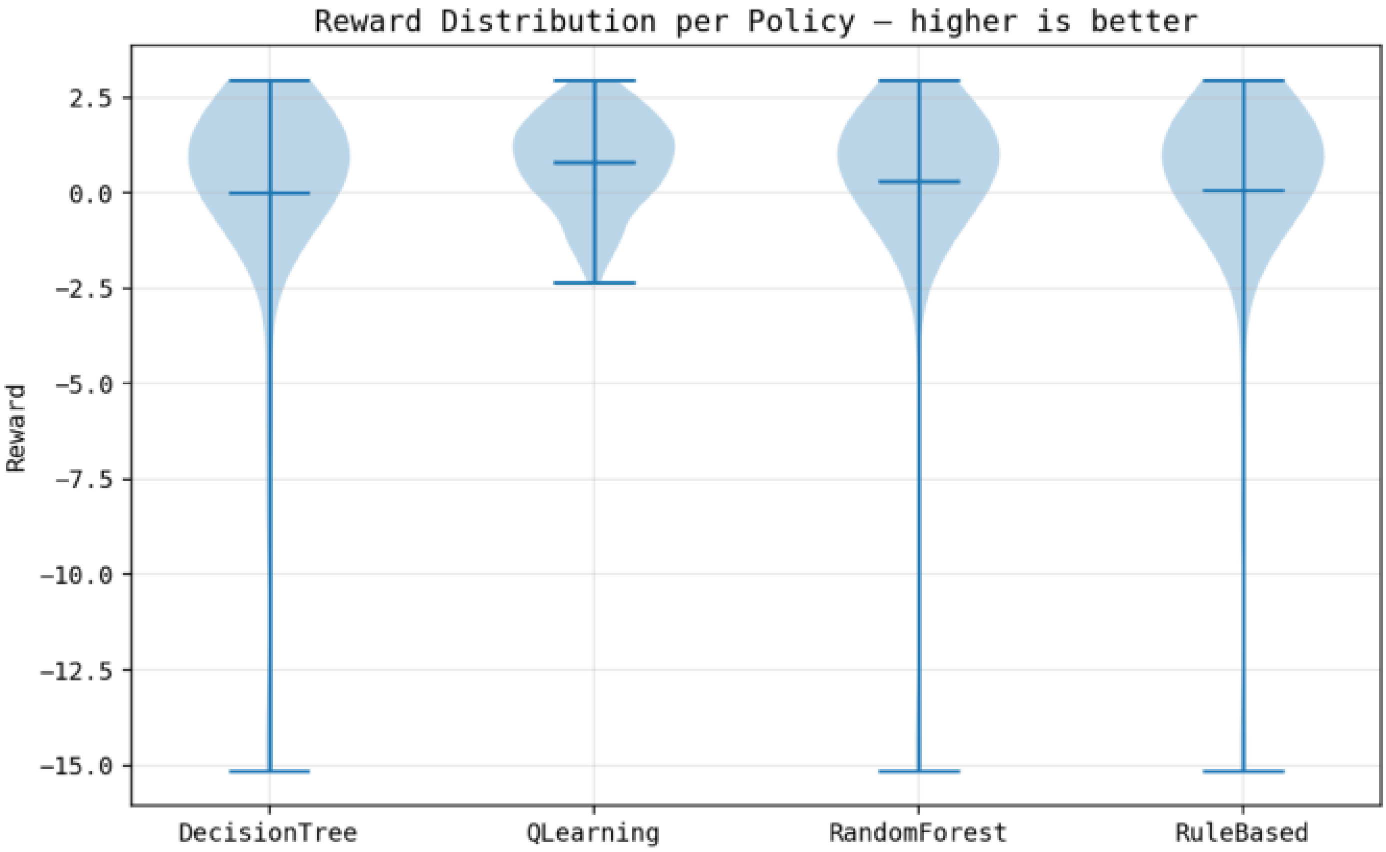
3.3 · Cumulative simulated cost



3.4 · Route distribution per policy



3.5 · Reward distribution



4 · Sample Explained Decisions

Example 1 — context:

network_latency_ms=56.7 · connectivity=1 · cpu_available=62.6 · memory_available=65.5 · batch_size=1 · priority=2 · cost_budget_usd=

· RuleBased → EDGE (conf 0.75, 9.2µs) net<80ms

· DecisionTree → EDGE (conf 1.00, 198.4µs) top-feature: network_latency_ms

· RandomForest → EDGE (conf 0.66, 11978.3µs) top-feature: cpu_available

· QLearning → EDGE (conf 1.00, 86.3µs) state=33 · Q=[1.56, -25.85, -8.09]

Example 2 — context:

network_latency_ms=116.7 · connectivity=1 · cpu_available=72.9 · memory_available=75.7 · batch_size=8 · priority=2 · cost_budget_usc

· RuleBased → EDGE (conf 0.85, 6.1µs) batch≥8

· DecisionTree → EDGE (conf 0.88, 190.9µs) top-feature: network_latency_ms

· RandomForest → EDGE (conf 0.84, 11963.7µs) top-feature: cpu_available

· QLearning → EDGE (conf 1.00, 80.1µs) state=69 · Q=[1.87, -32.54, -17.67]

Example 3 — context:

network_latency_ms=51.3 · connectivity=1 · cpu_available=52.9 · memory_available=48.9 · batch_size=1 · priority=3 · cost_budget_usd=

· RuleBased → EDGE (conf 0.75, 5.3µs) net<80ms

· DecisionTree → EDGE (conf 1.00, 388.4µs) top-feature: network_latency_ms

· RandomForest → EDGE (conf 0.68, 11643.1µs) top-feature: cpu_available

· QLearning → EDGE (conf 1.00, 81.0µs) state=21 · Q=[-0.05, -22.63, -6.92]

5 · Methodology & Reproducibility

1. Scenario definition
   Each experiment is executed under one of eight named scenarios
   (Mixed Traffic, Factory Normal, Network Failure, High CPU Load,
    Large Model, Cloud Congestion, Low Bandwidth, High Priority).

2. Reproducibility
   Given (scenario, seed), the platform produces the exact same context
   sequence, so all four policies are evaluated on IDENTICAL inputs
   (paired-samples design). Multiple seeds are used to compute
   confidence intervals via the Student t-distribution.

3. Metrics collected
   - Decision-time (µs) — orchestrator overhead
   - End-to-end latency (ms) — realised at the chosen route
   - Cost (USD)   — cumulative per policy
   - Route distribution — % of edge/cloud/hybrid selections
   - Success rate   — feasibility of the chosen route in context
   - Reward   — combined objective
   - Agreement w/ Rule   — divergence from the deterministic baseline

4. Threats to validity
   - Synthetic scenarios (mitigated by using diverse, semantically-named cases)
   - Q-Learning is single-step (contextual bandit): consistent with the
     routing problem, but not full episodic RL
   - Environment model estimates costs and latencies analytically

5. Artefact bundle
   /tmp/benchmark/{benchmark_records.csv, benchmark_summary.json,
                   benchmark_ci.json, chart_*.png, thesis_report.pdf}